Functional Annotation Report: Mouse *Lalba* (Alpha-lactalbumin, UniProt P29752)

Gene/Protein Identity Verification

The research target is unambiguously identified. UniProt **P29752** is **alpha-lactalbumin** (gene symbol **Lalba**, also "lactose synthase B protein") from *Mus musculus* (mouse). All verification criteria are satisfied:

- **Gene symbol match:** The literature on "Lalba"/"α-lactalbumin" describes exactly the protein in the UniProt description (the regulatory subunit of lactose synthase).
- **Organism match:** The definitive genetic studies were performed in mouse (e.g., the *Lalba*-null mouse,

  P 7708733

  ; isolation/sequencing of the mouse gene,

  P 1398111

).
- **Family/domain match:** Literature and the UniProt record both place the protein in the **glycosyl hydrolase 22 / c-type lysozyme superfamily** (Glyco_hydro_22, Lysozyme-like domain), consistent with the listed domains (IPR001916, IPR000545, IPR023346, PF00062).

There is **no ambiguity**: the symbol, organism, family, and domains are mutually consistent, and an extensive, high-quality primary literature exists for this exact protein.

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1. Summary (Answer to the Research Question)

Mouse alpha-lactalbumin (Lalba) is a small, secreted, calcium-binding protein that functions as the non-catalytic regulatory ("B" or modifier) subunit of the enzyme lactose synthase. It is not itself an enzyme. Its single, precise molecular function is to bind the

catalytic galactosyltransferase **β-1,4-galactosyltransferase-1 (β4GalT1)** and reprogram its sugar-acceptor specificity — switching the preferred acceptor from *N*-acetylglucosamine (GlcNAc) to free **glucose** — so that the complex synthesizes **lactose** (Galβ1-4Glc) from UDP-galactose and glucose. This reaction occurs in the **lumen of the Golgi apparatus of mammary secretory (alveolar) epithelial cells** during lactation; the protein is then packaged into secretory vesicles and exported as an abundant whey protein of milk. Because lactose is the principal osmole of milk, α-lactalbumin — by enabling lactose synthesis — effectively sets the **osmotic water content and volume of milk**, making it essential for successful lactation. Its expression is restricted to differentiated lactating mammary epithelium and is driven by the **prolactin → PRLR → JAK2/STAT5 lactogenic signaling pathway**.

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2. Primary Molecular Function: The Specifier Subunit of Lactose Synthase

2.1 The reaction catalyzed and the role of Lalba

Lactose synthase (EC 2.4.1.22) is a **1:1 heterodimeric complex** of two components:

- the **catalytic "A protein"**, β-1,4-galactosyltransferase-1 (β4GalT1), a Mn^{2+}-dependent, trans-Golgi, type II membrane glycosyltransferase; and
- the **regulatory "B protein"**, α-lactalbumin (the product of *Lalba*).

On its own, β4GalT1 transfers galactose from UDP-galactose onto GlcNAc residues of glycoconjugates (forming the Galβ1-4GlcNAc unit, *N*-acetyllactosamine). **α-Lactalbumin is a non-catalytic specifier**: when it binds β4GalT1 it promotes the binding of free glucose and alters the enzyme's acceptor specificity from GlcNAc to glucose, enabling synthesis of lactose — the defining carbohydrate of milk (Qasba et al., *Curr Drug Targets* 2008,

P 18393823

; Ramakrishnan & Qasba 2001,

P 11419947

).

As a 2012 review states, α-lactalbumin "in its native state acts as a substrate specifier in the

lactose synthase complex, thereby defining a function essential for the survival of lactating mammals" (Ho et al. 2012,).

P 23130929

- **Substrates of the α-lactalbumin-modified reaction:** UDP-galactose (sugar donor) + D-glucose (acceptor); requires Mn^{2+}.
- **Product:** lactose (Galβ1-4Glc) + UDP.
- **Substrate-specificity change conferred by Lalba:** acceptor shifts from GlcNAc → glucose (dramatic lowering of the apparent K_m for glucose).

2.2 Structural mechanism (precise, crystallographic evidence)

The 2 Å crystal structures of lactose synthase reveal the physical basis of the specifier function (Ramakrishnan & Qasba 2001,).

P 11419947

Substrate binding triggers a **large conformational change in β4GalT1 (residues ~345–365)**, repositioning His347 to coordinate the catalytic metal ion and simultaneously **creating the α-lactalbumin- and sugar-acceptor-binding site**. α-Lactalbumin then **holds glucose in place by hydrogen bonding to its O-1 hydroxyl group** in the acceptor site; a reorientation of Arg359 blocks the hydrophobic *N*-acetyl pocket (which normally accommodates the GlcNAc *N*-acetyl group) and maximizes interactions with glucose. The direct enzyme–modifier complex has been captured crystallographically, including a pentenary complex of bovine Gal-T1·Mn^{2+}·UDP-GalNAc·Glc·α-lactalbumin (Ramakrishnan et al. 2006,).

P 16497331

Comparative crystal structures of species variants show that the functionally important region (mature residues ~105–110), adjacent to the monosaccharide-binding residues Phe31/His32, is conformationally flexible and forms the enzyme-interaction surface (Pike et al. 1996,).

P 8805552

2.3 Kinetic mechanism

Classical enzyme kinetics established that lactose synthesis proceeds by a **random-equilibrium mechanism**: binding of Mn^{2+} at the tight metal site (Kd ≈ 2.3 μM) is an obligatory prerequisite, and α-lactalbumin can associate with either the enzyme·Mn^{2+}·acceptor or the enzyme·Mn^{2+}·UDP-galactose complex (Bell et al. 1976,

; P 818085 Powell & Brew 1976,). P 932002

Thus α-lactalbumin does not change the catalytic chemistry; it modulates acceptor binding.

Interpretation: The primary function of Lalba is best described as a **substrate-specificity modifier / regulatory binding subunit**, converting a general glycosyltransferase into the dedicated lactose-synthesizing enzyme of lactation.

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3. Subcellular and Extracellular Localization

- **Site of action (intracellular):** the reaction occurs in the **lumen of the Golgi apparatus** of mammary epithelial cells, where UDP-galactose is available. Lactose "is synthesized in the Golgi of mammary epithelial cells by lactose synthase from glucose and UDP galactose," and α-lactalbumin co-localizes with Golgi markers (58K protein, GM130) and with the GLUT8 glucose transporter at the cis-Golgi lactose-synthase complex (Villagrán et al. 2019,). P 31020623 Its catalytic partner β4GalT1 is a trans-Golgi type II membrane protein ( ). P 18393823
- **Secretory routing:** α-Lactalbumin is synthesized as a **precursor with a cleaved N-terminal signal peptide** (UniProt P29752: signal 1–20; secreted mature chain 21–143), directing it through the ER → Golgi secretory pathway. Pulse-chase studies show α-lactalbumin is sequestered within membrane-bound secretory vesicles and secreted ~30 min after synthesis (Boulton et al. 1984,). P 6477529
- **Extracellular destination:** It is ultimately **secreted into milk** as one of the major whey proteins (~0.8 mg/ml in mouse milk;). P 10085224

Thus Lalba acts **intracellularly (Golgi lumen)** as the specifier subunit and is also **exported extracellularly** into milk.

4. Biological Process / Pathway Context

4.1 Lactose biosynthesis and the osmotic control of milk volume

Lactose is the **major osmole of milk**. Because the Golgi membrane is relatively impermeant to lactose, **intravesicular lactose synthesis osmotically draws water into Golgi-derived secretory vesicles, causing them to swell before exocytosis** — directly coupling the rate of lactose production to the aqueous volume (yield) of milk (Holt 1983, P 6876831).

This mechanistic link explains why *Lalba* ultimately governs milk volume.

4.2 Genetic proof of the primary function (mouse knockout)

Targeted deletion of the mouse *Lalba* gene provides definitive in vivo evidence: **homozygous null females are α-lactalbumin-deficient, produce reduced amounts of thickened (highly viscous) milk containing little or no lactose, and cannot sustain their offspring** (Stacey et al. 1995, P 7708733).

Replacement with the human α-lac gene (humanized allele) restores normal lactation, and expression is gene-dosage dependent, with increased expression slightly increasing milk volume (P 7708733).

An independent α-lac-deficient line likewise produces **lactose-free, viscous milk with a shortened lactation period** (Soulier et al. 1999, P 10095792), and a bovine α-lac transgene rescues lactation in deficient mice (P 9120073).

These results establish that α-lactalbumin is **required for lactose synthesis** and that lactose is essential for normal milk production and offspring survival.

4.3 Upstream regulation: lactogenic hormone signaling

Lalba is a **committed, hormonally gated effector** of lactation rather than a signaling molecule. Its transcription is strongly induced during functional differentiation of the mammary gland by **lactogenic hormones — principally prolactin acting through the prolactin receptor (PRLR) and the JAK2/STAT5 pathway** (together with insulin and glucocorticoid):

- Insulin + hydrocortisone + prolactin induce α-lactalbumin mRNA several-hundred-fold in mammary explants (Shao et al. 2013, P 23063409).
- Prolactin induces α-lactalbumin dose-dependently in mammary epithelial cells (P 26641038).
- Conditional repression of STAT5 during lactation reduces α-lactalbumin expression and slows neonate growth (Reichenstein et al. 2011, P 21688337).

Accordingly, Lalba sits **downstream** of the prolactin–PRLR–STAT5 program as its biochemical output.

4.4 Substrate supply

Glucose supply to the Golgi lactose-synthase complex is provided by glucose transporters (GLUT1 and GLUT8), which are co-regulated with lactation and localize with the lactose-synthase machinery (P 31020623).

UDP-galactose is generated by the Leloir pathway and imported into the Golgi; its supply can be rate-limiting for lactose synthesis (Mohammad et al. 2012, P 22649065).

5. Structure, Evolution, and Mouse-Sequence Evidence

5.1 A calcium-binding, lysozyme-fold protein that lost muramidase activity

α-Lactalbumin belongs to the **c-type lysozyme superfamily (glycosyl hydrolase 22 family)** and adopts the lysozyme fold, yet it has **no muramidase (glycosidase) activity** — it was evolutionarily repurposed from an ancestral lysozyme by gene duplication within the mammalian lysozyme gene family, acquiring a divergent regulatory function (Irwin et al. 2011, P 21676251; Prager & Jollès 1996, P 8765292).

High-resolution X-ray structures show it binds a **single Ca^{2+} ion in a "novel" calcium-binding loop** (slightly distorted pentagonal-bipyramidal coordination, Ca–O distances 2.2–2.6 Å) that superficially resembles an EF-hand but is instead derived from the homologous loop of c-type lysozyme (Stuart et al. 1986, P 3785375; Acharya et al. 1989, P 2769757).

Ca^{2+} binding stabilizes the native fold required for productive interaction with β4GalT1; conversely, loss of Ca^{2+} promotes the partially unfolded (molten-globule) state.

5.2 Direct confirmation from the mouse P29752 sequence

Analysis of the mouse UniProt record (143-aa precursor) independently confirms all functional hallmarks at the mouse-sequence level:

- **Signal peptide (residues 1–20)** cleaved to a **123-aa secreted mature chain (21–143)**, annotated as a C-type lysozyme domain — consistent with Golgi routing and milk secretion.
- **Ca^{2+}-binding loop** formed by mature residues **Lys79, Asp82, Asp87, Asp88** (the aspartate-rich "DKLLDDELDDD" motif) — the canonical α-lactalbumin calcium loop.
- **Four disulfide bonds** (mature Cys6–Cys120, Cys28–Cys111, Cys61–Cys77, Cys73–Cys91) matching the conserved c-type lysozyme disulfide pattern.
- **One N-glycosylation site** (Asn65 in precursor numbering).

- Secondary-structure annotation shows the expected **α+β lysozyme fold** (four α-helices plus a small β-sheet).
- The functionally critical **monosaccharide (glucose)-binding residues Phe31 and His32** (mature numbering) implicated in lactose synthase (P 8805552) are **conserved in the mouse sequence**, directly linking the mouse ortholog to the glucose-binding/specifier mechanism derived from other mammalian structures.

These features corroborate, at the level of the actual mouse protein, the structural/mechanistic roles established from bovine/human/baboon crystal structures.

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6. A Note on Moonlighting Activity (HAMLET) — Not the Primary Function

When α-lactalbumin **partially unfolds** (e.g., upon Ca^{2+} release and low pH) and binds **oleic acid**, it forms **HAMLET** (Human Alpha-lactalbumin Made LEthal to Tumor cells), a protein–lipid complex selectively toxic to tumor cells (Ho et al. 2012, P 23130929 ; Nadeem et al. 2019, P 31082436). Evidence indicates the **oleic acid cofactor is the principal cytotoxic agent**, with the protein acting as a solubilizing carrier (El-Fakharany & Redwan 2019, P 35539089 ; Hoque et al. 2013, P 24039698). This is a **gain-of-function of a conformationally altered, non-native state** — not the physiological function of Lalba — and is most relevant to human/bovine α-lactalbumin as a therapeutic. It is included here only for completeness and to avoid conflating it with the primary lactose-synthase specifier role.

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7. Supported and Refuted Hypotheses

Supported (high confidence, multiple independent lines of evidence): - H1. Lalba is the non-catalytic regulatory subunit of lactose synthase. — *Supported* (crystallography, kinetics, biochemistry; PMIDs 11419947, 18393823, 818085). - H2. Lalba changes β4GalT1 acceptor specificity from GlcNAc to glucose to make lactose. — *Supported* (PMIDs 11419947, 16497331). - H3. Lalba function/site of action is the Golgi lumen of mammary epithelial cells; the protein is secreted into milk. — *Supported* (PMIDs 31020623, 6477529; UniProt signal peptide). - H4. Lalba is genetically required for milk lactose and successful lactation. — *Supported* (mouse knockout; PMIDs 7708733, 10095792, 9120073). - H5. Lactose (enabled by Lalba) osmotically determines milk volume. — *Supported* (PMIDs 6876831, 7708733). - H6. Lalba expression is controlled by prolactin/STAT5 lactogenic signaling. — *Supported* (PMIDs 23063409, 21688337, 26641038). - H7. Lalba is a Ca^{2+}-binding lysozyme-fold protein without muramidase activity. — *Supported* (PMIDs 3785375, 21676251; mouse sequence analysis).

Refuted / excluded: - Lalba is itself a catalytic glycosyltransferase/hydrolase. — *Refuted*: it is non-catalytic and lacks the muramidase activity of its lysozyme ancestor (PMIDs 18393823, 21676251). - HAMLET tumoricidal activity is the native cellular function of Lalba. — *Refuted/clarified*: it requires non-native partial unfolding + oleic acid and is not the physiological role (PMIDs 23130929, 35539089).

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8. Limitations and Future Directions

- Several atomic-resolution mechanistic studies (crystallography, kinetics) were performed with **bovine, human, or baboon** α-lactalbumin; the mouse ortholog is highly conserved and the mouse genetic data (knockout) are definitive, but a mouse-specific lactose-synthase crystal structure is not the basis of the mechanistic claims. The mouse UniProt sequence analysis directly confirms conservation of the functional residues.
- Quantitative mouse-specific kinetic constants (K_m for glucose with/without α-lactalbumin) are drawn from bovine enzymology.
- Fine regulation (e.g., the precise transcription-factor complement at the mouse "milk-box"/STAT5 elements) and any post-secretory roles of milk α-lactalbumin in the neonate were not exhaustively surveyed.

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9. Key References (PMIDs)

- 11419947 — Ramakrishnan & Qasba, crystal structure of lactose synthase (specificity switch mechanism).
- 18393823 — Qasba et al., structure/function of β-1,4-galactosyltransferase and α-lactalbumin.
- 16497331 — Ramakrishnan et al., structural snapshots / pentenary complex with α-lactalbumin.
- 818085 / 932002 — Bell et al.; Powell & Brew, kinetic mechanism and metal/α-lactalbumin binding.
- 7708733 — Stacey et al., *Lalba*-null and humanized mice (definitive genetic proof).
- 10095792 / 9120073 — Soulier et al., α-lac-deficient mice; bovine transgene rescue.
- 31020623 — Villagrán et al., Golgi localization and glucose transporters.
- 6477529 — Boulton et al., secretory-pathway transit of α-lactalbumin.
- 6876831 — Holt, lactose-driven osmotic swelling of Golgi vesicles / milk volume.
- 23063409 / 21688337 / 26641038 — lactogenic hormone (prolactin/STAT5) induction of α-lactalbumin.
- 1398111 / 2845947 / 10085224 — mouse gene isolation; milk-box promoter; expression level.
- 3785375 / 2769757 / 8805552 — α-lactalbumin calcium loop and lysozyme-fold structures.
- 21676251 / 8765292 — evolution within the lysozyme gene family.
- 23130929 / 31082436 / 35539089 / 24039698 / 34103518 — HAMLET moonlighting activity (non-native).